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Major Project/Internship

1010043492

HireHub

Name : Berwa Harsh Govindbhai

Enrollment No : 2301031030003

Branch: Information Technology

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Introduction

HireHub is a web-based job portal application developed using the MERN stack, which includes MongoDB, Express.js, React.js, and Node.js. The application provides a centralized platform where students can search for job opportunities and apply for them, while HR professionals can post job openings and manage applicants efficiently. The system is designed with a focus on simplicity, usability, and smooth interaction between users, making the hiring process more organized and accessible.

Background and Motivation

In today's digital environment, job searching and recruitment processes are often distributed across multiple platforms, which makes it difficult for students to find relevant opportunities and for HR to manage applications effectively. Students frequently face confusion due to scattered job listings, while HR has to manually filter and track numerous applications. This situation created the need for a centralized and easy-to-use system, which motivated the development of HireHub to simplify and streamline the hiring process.

Relevance and Importance

HireHub is important because it reduces manual effort in the recruitment process and provides a single platform for both job seekers and recruiters. It helps in improving efficiency, saving time, and maintaining transparency in hiring. The system is especially useful for students, educational institutions, and small organizations that require a simple and effective job management solution. By integrating all essential features in one place, HireHub enhances the overall user experience.

Literature review

Existing job platforms such as LinkedIn, Indeed, and Naukri offer a wide range of services, but they are often complex and include many advanced features that may not be necessary for all users. HireHub takes inspiration from these platforms but focuses on providing only the core functionalities such as job posting, job application, and application tracking. This makes the system lightweight, user-friendly, and easy to understand, especially for beginners and students.

Objectives

The main objective of this project is to develop a full-stack job portal using the MERN stack that supports secure and efficient interaction between students and HR. The system aims to implement role-based authentication to differentiate between users and provide specific functionalities such as job application for students and applicant management for HR. It also focuses on preventing duplicate applications, maintaining data consistency, and ensuring a smooth user experience.

Required Tools and Technology

The project is developed using modern web technologies. The frontend is built using React.js for creating dynamic user interfaces and Tailwind CSS for styling and responsiveness. The backend is developed using Node.js and Express.js to handle server-side logic and API creation. MongoDB is used as the database to store user, job, and application data. Additional tools include JSON Web Token (JWT) for secure authentication, Postman for API testing, and GitHub for version control and project management.

Method/Approach

The development of HireHub followed a structured and modular approach. Initially, the backend was developed by creating RESTful APIs for user authentication, job posting, and application management, with MongoDB used for data storage and relationships. The frontend was then built using React.js with components for login, job listing, and applicant management, styled using Tailwind CSS for a responsive UI. JWT authentication was implemented to secure the application and manage role-based access. Finally, the frontend and backend were integrated, and features such as job application, status updates, and protected routes were tested and optimized.

Bibliography

The development of this project was supported by various online resources and official documentation. These include the official documentation of MongoDB, React.js, Node.js, Express.js, and Tailwind CSS, which provided detailed guidance on implementation. Additional references include MDN Web Docs for web development concepts and various YouTube tutorials and online learning platforms that helped in understanding practical implementation and problem-solving during the project development.